

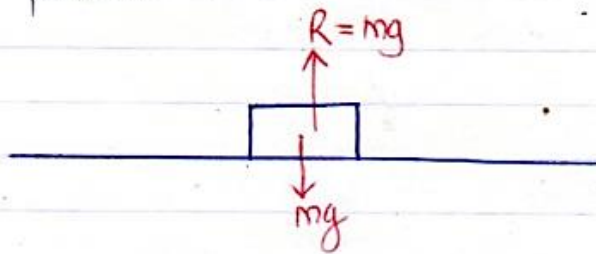
Topic 6 Summary: Friction

- Definition and Explanation of Frictional force.
- Static Friction
 - Description and explanation
 - Maximum value of static friction/formula
 - Coefficient of static friction
- Kinetic Friction
 - Description and explanation
 - Formula
 - Coefficient of kinetic friction
- Relationship with Normal force
- What does μ depend on?
- Measuring μ and the angle of friction
- Worked examples

Force and Newton's Laws : Friction

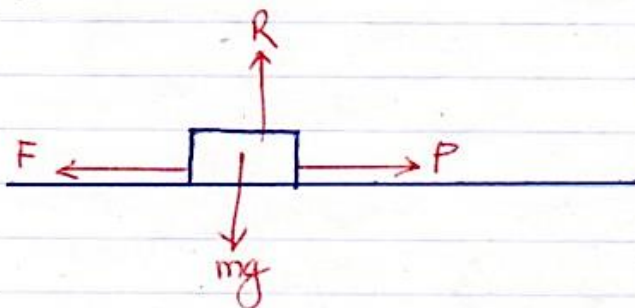
The force of friction opposes motion when one surface slides over another. It also plays an important role in the equilibrium of stationary bodies.

Consider an object of mass m lying on a rough horizontal plane.



If the object is at rest (equilibrium), then the only forces acting are the weight (mg) and the normal contact force (R). There is no force of friction.

If a small force P is now applied to the object horizontally but is insufficient to cause motion, then to maintain equilibrium, there must be an equal and opposite force to P . This force is supplied by **friction (F)**, and is called **Static Friction**.



If P is 1N , then F will be 1N ; if $P = 10\text{N}$, F will be 10N . If P is removed, the frictional force F will vanish as well.

If the force P continues to increase, eventually the object will move (slide). Suppose this occurs when

$P = 12\text{ N}$. The ^{static} frictional force F will also have increased to 12 N or almost 12 N , but is no longer able to balance the force P . i.e. it has reached its maximum value.

If P continues to increase, F will remain at its maximum value of 12 N and the object will now accelerate under the resultant force $(P - 12)\text{ N}$.

Thus we see that the magnitude of static friction can vary from zero to some maximum value when the object begins to slip.

The magnitude of ^{the maximum value} static friction is proportional to the magnitude of the normal reaction R . i.e.

$$F_{\max} = \mu_s R$$

Here the constant of proportionality, μ_s , is the **coefficient of static friction**.

There is also another type of friction called **kinetic friction** or **sliding friction**. This occurs when two objects in contact are moving relative to each other.

Based on experimental measurements, the force of kinetic friction between two surfaces, is independent of the relative speed of the surfaces, the area of contact, the shape of the objects etc, and only depends on the magnitude of the normal reaction R . i.e.

$$F = \mu_k R$$

where μ_k is called the coefficient of kinetic friction.

$$\mu_s > \mu_k$$

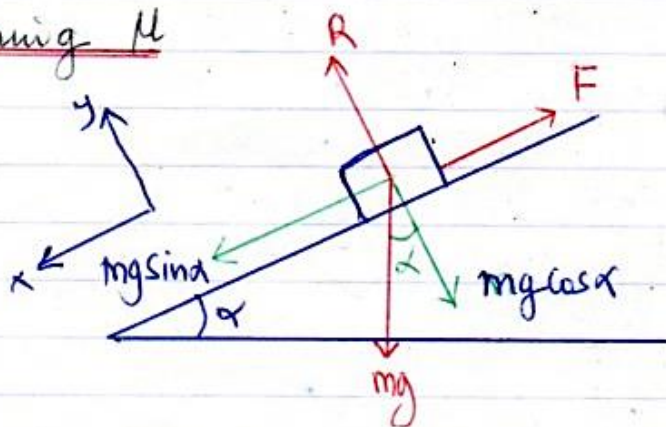
The coefficient of static friction is usually slightly greater than that of kinetic friction.

In practice, when motion between two bodies in contact takes place, the force of kinetic friction is slightly less than that of static friction. When great accuracy is required, μ_s and μ_k are both quoted for the same surface. However, in most cases, it is sufficient to regard the two as equal and simply refer to it as the coefficient of friction.

The coefficient of friction depends solely upon the nature of the surfaces in contact, and NOT on their relative masses or surface area of contact.

e.g. for two smooth sheets of glass, $\mu = 0.05$
for a tyre & tarmacked road, $\mu \approx 1$

Measuring μ



When an object rests on an incline, the force of friction F prevents it from sliding.

As the tilt is increased, i.e. as α increases, F becomes larger and larger until it reaches its max value and the object is about to slide.

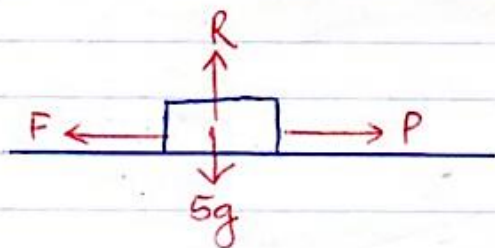
At that point

$$\begin{aligned}mg \sin \alpha - F_{\max} &= 0 \\mg \sin \alpha - \mu_s mg \cos \alpha &= 0 \\ \Rightarrow \boxed{\mu_s = \tan \alpha}\end{aligned}$$

Example 1:

A block of mass 5kg rests on a rough horizontal plane, the coefficient of friction between the block and the plane being 0.6. Calculate the frictional force acting on the block when a horizontal force P is applied to the block and the magnitude of P is
a) 12 N b) 28 N c) 36 N.

Sol: ..



In y-direction:
 $R = 5g$

Frictional force will act opposite to P . Max value of F
 $F_{\max} = \mu R = 0.6 \times 5 \times 9.81 = 29.43 \text{ N}.$

- a) If $P = 12 \text{ N}$, then P is less than $\mu R \therefore$ block will not move
 $F = P = \underline{12 \text{ N}}$
- b) If $P = 28 \text{ N}$, again less than $\mu R \therefore$ no motion
 $F = P = \underline{28 \text{ N}}.$
- c) If $P = 36 \text{ N}$, then $P > \mu R$, block will move and
 $F = \mu R = \underline{29.4 \text{ N}}$

Example 2:-

A crate of mass 20kg lies on a rough horizontal floor, the coefficient of friction being 0.7. Find:

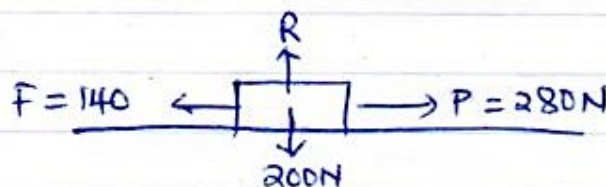
- the minimum horizontal force needed to move the crate
- the acceleration if twice the minimum force is used

Sol:

a) Static friction $F = \mu R$
 $= 0.7 \times 20 \times 10$
 $= 140 \text{ N.}$

The minimum force P that will just move the crate is 140 N.

b) Force applied $= 2 \times 140 = 280 \text{ N}$



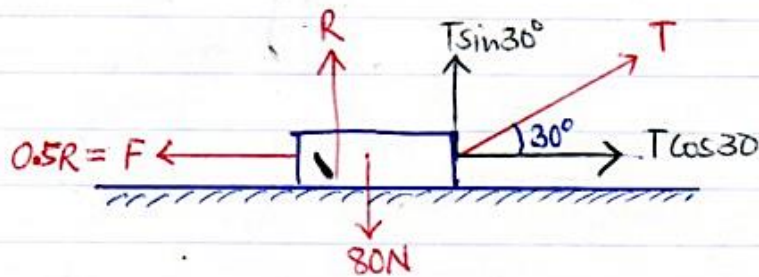
Resultant force $= P - F$
 $= 280 - 140 = 140 \text{ N}$

$F = ma$
 $140 = 20a \Rightarrow a = \underline{\underline{7 \text{ m/s}^2}}$

Example 3

A box of mass 8 kg, standing on rough horizontal ground, is pulled by a string inclined at 30° to the horizontal. If the body is about to slide and $\mu = 0.5$, find the tension in the string.

Sol:



In y-direction

$$R + T \sin 30 = 80 \text{ N} \quad \text{--- (i)}$$

In x-direction

$$T \cos 30 - 0.5R = 0 \quad \text{--- (ii)}$$

$$\Rightarrow R = 2T \cos 30$$

and $\therefore$ (i) becomes

$$2T \cos 30 + T \sin 30 = 80$$

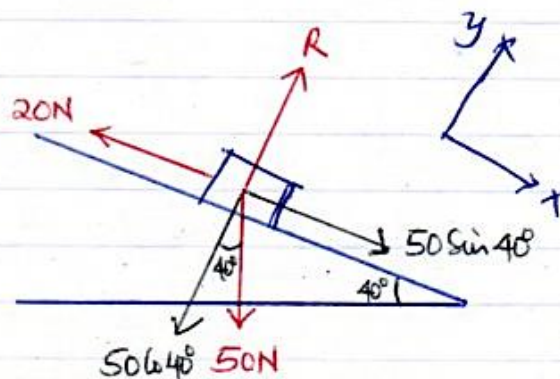
$$T(1.732 + 0.5) = 80$$

$$T = \underline{\underline{35.84 \text{ N}}}$$

Example 4: E.g 6 Dyn.

A body of mass 5kg is allowed to slide down a rough plane inclined at 40° to the horizontal, against a frictional resistance of 20N. Find the acceleration $\vec{a}$ down the plane and the normal contact force R.

Sol:



In the y-direction

$$R = 50 \cos 40 \\ = \underline{\underline{38.302 \text{ N}}}$$

In the x-direction

$$50 \sin 40 - 20 = ma$$

$$50 \sin 40 - 20 = 5a$$

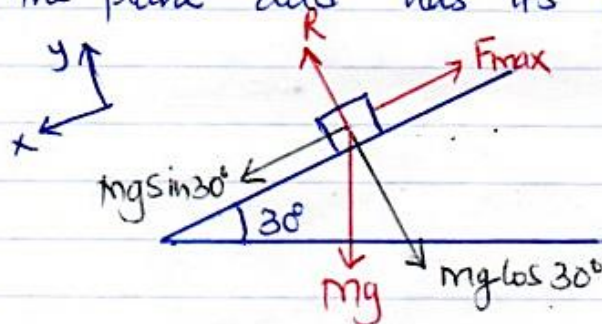
$$\Rightarrow a = \underline{\underline{2.428 \text{ m/s}^2}}$$

Example 5

A mass of 6 kg is at the point of moving on a rough plane inclined at 30° to the horizontal. Find the coefficient of friction between the mass and the plane.

Sol:

Mass is at the point of moving down, so F acts up the plane and has its maximum value $\mu_s R$.



In y-direction

$$R = mg \cos 30^\circ = 6g \cos 30 = 50.97 \text{ N.}$$

In x-dir

$$mg \sin 30 - F_{\max} = 0$$

$$mg \sin 30 - \mu_s mg \cos 30 = 0$$

$$\mu_s = \frac{\sin 30}{\cos 30} = \tan 30^\circ = \underline{\underline{\frac{1}{\sqrt{3}}}} = 0.58$$

Example 6

A mass of 0.5 kg rests on a rough plane. The coefficient of kinetic friction between the mass and the plane is $\frac{1}{\sqrt{2}}$ and the plane is inclined at the angle α to the horizontal such that

$\sin \alpha = \frac{1}{3}$. Investigate the motion of the mass when it experiences a force of 6 N applied up the plane along the line of greatest slope, and hence find the acceleration.

Ans: $\vec{a} = 2.19 \text{ ms}^{-2}$